

UNIVERSITY OF PLYMOUTH

BSC (HONS) COMPUTER SCIENCE (SOFTWARE ENGINEERING)

MAL3018 COMPUTING PROJECT

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Project Name : Wearly: An AI-Based Personal Wardrobe and Outfit Recommendation System

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1.0 Project Title

Wearly: An AI-Based Personal Wardrobe and Outfit Recommendation System

2.0 Keywords

Artificial Intelligence, Computer Vision, Digital Wardrobe, Outfit Recommendation, Virtual Try-On, FashionCLIP, Clothing Image Analysis, Personalised Recommendation

3.0 Links

GitHub: https://github.com/YEEPEI061/Wearly.git

Pebblepad: https://v3.pebblepad.co.uk/spa/#/public/mtw79ZdGhWkw7j4b87c5z85hdw

4.0 Student Information

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6.0 Background and Problem Statement

Background

The increased availability of clothing and frequent changes in fashion trends have resulted in greater personal wardrobes; however, many clothing items may remain underutilised. At the same time, deciding what to wear every day can be time-consuming, especially if users have a lot of items but aren't sure how to blend them into appropriate outfits. This can make it difficult for users to make full use of their existing clothing. Making better use of current garments can help to promote more sustainable clothing consumption by encouraging people to use their clothes for longer and more frequently, rather than purchasing or disposing of them prematurely.

At the same time, artificial intelligence and computer vision have created opportunities for intelligent fashion applications. AI-based clothing analysis and outfit recommendation can generate suitable outfit combinations from items available in a user's personal wardrobe, while virtual try-on allows users to visualise how selected clothing may look when worn before making an outfit decision.

However, current fashion applications often focus on recommending items for purchase, clothing from predefined catalogues, or social and community-based fashion features. There is potential for a more personalised system that learns a user's wardrobe, analyses specific clothing items, and offers outfit recommendations based on the user's tastes. Such a system might also serve as a foundation for selecting appropriate wardrobe additions as needed, while encouraging people to make better use of their existing clothing.

Problem Statement

Users may own a large number of clothing items but still experience difficulty deciding what to wear or identifying different combinations from their existing wardrobe. Manually organising clothing and determining suitable combinations can also be time-consuming.

Existing fashion recommendation approaches can make clothing recommendations; however, many are based on commercial fashion catalogues or internet buying rather than the user's personal wardrobe. Research has shown that outfit recommendation from a personal closet is technically feasible, but there remains an opportunity to combine personal wardrobe management, clothing image analysis, outfit recommendation, virtual try-on, and user feedback-based personalisation into a single mobile application.

Therefore, this project proposes **Wearly**, an AI-powered mobile application that analyses users' clothing images, organises them into a digital wardrobe, and generates outfit combinations based on clothing compatibility. The system also incorporates like/dislike feedback to learn users' preferences and improve subsequent recommendations.

The project supports **SDG 9: Industry, Innovation and Infrastructure** by applying artificial intelligence and computer vision to develop an intelligent system for clothing analysis, wardrobe management, and personalised outfit recommendation. It also aligns with **SDG 12: Responsible Consumption and Production** by encouraging people to make better use of clothing they already own. Clothing reuse and extended usage can help users maximise the value of their existing garments, resulting in more sustainable consumption.

Target User

The primary target users are individuals who often ask themselves, “What should I wear today?” and want an easier way to organise their wardrobe and discover suitable outfit combinations.

7.0 Objectives

- **To design and develop a mobile-based digital wardrobe application** that allows users to upload or capture clothing images and organise clothing items into categories.
- **To implement AI-based clothing image analysis and outfit recommendation** using clothing segmentation, visual feature extraction, image classification, and compatibility analysis to identify clothing attributes and generate suitable outfit combinations.
- **To implement a feedback-based personalisation mechanism** that uses users' like/dislike feedback to learn their clothing preferences and adjust future outfit recommendations.
- **To evaluate the effectiveness and usability of the proposed system** through functional testing, recommendation evaluation, and user testing.

8.0 Scope of the Project

Digital Wardrobe Management

Users will be able to upload or capture images of their clothing items through the mobile application. The system will process the images and store the resulting clothing information in a digital wardrobe.

The initial system will support major clothing categories, including:

- Tops
- Bottoms
- Full-body garments

AI-Based Clothing Analysis

The system will use AI-based computer vision techniques to analyse clothing images.

The proposed implementation includes:

- **FASHN Human Parser** for clothing segmentation and extraction
- **CLIP** for colour and style classification
- **FashionCLIP** for fashion-specific visual feature extraction and similarity analysis

The extracted clothing information and visual features will be used to support clothing organisation, outfit compatibility analysis, and personalised outfit recommendations.

AI-Assisted Outfit Recommendation

The system will generate outfit combinations from clothing items available in the user's wardrobe.

The recommendation process will consider:

- Clothing category compatibility
- Colour compatibility
- Style compatibility
- Visual similarity
- Previously generated outfits

Personalised Outfit Recommendation

Users will be able to provide **Like/Dislike feedback** on generated outfits.

The system will use this feedback to build a preference profile based on attributes such as:

- Preferred colours
- Preferred styles
- Clothing categories
- Visual similarity using FashionCLIP features

The preference profile will then influence the ranking of future outfit recommendations.

Outfit History

The system will maintain a history of outfits generated and selected by the user. A calendar-based interface will allow users to view their previous outfits by date, helping them keep track of what they have worn and discover outfit combinations from their past. Users can also revisit previous outfit combinations and use them as references when planning future outfits.

AI Virtual Try-On

The system will provide an AI-based virtual try-on feature that allows users to preview selected clothing items on their own images. Users will be able to upload or capture a suitable personal image and select a clothing item from their digital wardrobe. Users will be able to upload or capture an appropriate personal image and select a recommended outfit to preview on themselves.

The initial implementation will focus on front-facing images and selected outfits, while additional views such as back-facing images may be considered depending on the available AI technology and development resources.

Sustainable Wardrobe Utilisation

The application will encourage users to explore different combinations using clothing already available in their wardrobe.

This supports **SDG 12: Responsible Consumption and Production** by promoting better utilisation and reuse of existing clothing rather than encouraging unnecessary replacement or purchasing.

Mobile Application

The project will be developed as a mobile application with functions for:

- Clothing upload or capture
- Digital wardrobe management
- AI clothing analysis
- Outfit generation
- Like/dislike feedback
- Personalised recommendations
- Outfit history
- AI virtual try-on

Out of Scope

To maintain a manageable scope for the initial implementation, the following functions will not be included in the core FYP implementation:

- Direct clothing purchasing
- E-commerce integration
- Payment functionality
- Automatic clothing disposal or recycling services

9.0 Literature Review / Related Works

9.1 Existing Digital Wardrobe and Fashion Recommendation Applications

Feature	Acloset	Whering	Cladwell	Proposed System
Digital Wardrobe	✓	✓	✓	✓
AI Clothing Analysis	✓	Limited	Limited	✓
AI Outfit Recommendation	✓	✓	✓	✓
User Preference Personalisation	✓	✗	✓	✓ (Feedback-based)
Like/Dislike Feedback Learning	Limited / not central	✗	Limited	✓
Outfit History / Calendar	✓	✓	✓	✓
Virtual Try-On	✓	✗	✗	✓
Travel Outfit Planning	✓	✓	✓	✓

Main Focus	AI wardrobe + social fashion	Social wardrobe + planning	Personal styling	**AI wardrobe + personalised styling**

Application	**Main Strength**	**Limitation / Gap**
Acloset	Combines digital wardrobe, AI styling, weather-based recommendations, and virtual try-on.	Provides a wide range of AI-powered wardrobe and styling features. However, its recommendations may include items beyond the user's existing wardrobe.
Whering	Strong wardrobe organisation, outfit planning, and social fashion interaction.	Strongly oriented toward social/community styling and user-created outfit planning rather than AI-driven clothing analysis as the central function.
Cladwell	Focuses on personalised daily styling, weather, and wardrobe planning.	Does not provide virtual try-on and has a stronger capsule wardrobe and personal-styling approach.
Proposed System	Combines AI clothing analysis, outfit recommendation, feedback-based personalisation, wardrobe	Focuses on personal wardrobe assistance rather than social or community interaction.

	history, and virtual try-on in one system.	

Existing apps offer various mixes of digital wardrobe management, outfit planning, AI recommendation, weather-based style, social interaction, and virtual try-ons. The suggested solution combines AI clothing analysis, personalised outfit recommendations based on explicit user feedback, outfit history, and virtual try-on into a single personal wardrobe application without making social networking a primary purpose.

9.2 AI-Based Outfit Recommendation

Artificial intelligence is increasingly being used to recommend outfits based on clothing attributes and compatibility between different clothing items. A deep neural network has been proposed to evaluate outfit combinations from a user's personal closet, demonstrating the feasibility of generating suitable outfits from uncurated wardrobe items (Tangseng et al., 2018).

Deep learning and computer vision have also been explored for clothing feature extraction and recommendation, showing that visual features can be used to model relationships between clothing items (Liu et al., 2022). Multi-modal deep learning approaches have further combined visual and textual information to improve clothing compatibility analysis, demonstrating the importance of extracting meaningful clothing features when generating outfit recommendations (Sohn et al., 2026).

Personalisation has also been explored in outfit recommendations. A style-guided multi-modal approach, POSM, incorporates personalised style information into outfit recommendation, highlighting the importance of considering individual preferences rather than relying only on general clothing compatibility (You et al., 2025).

Wearly is built on these findings, combining AI-based garment analysis, visual feature extraction, outfit compatibility analysis, and feedback-based personalisation. The system uses the analysed clothing features and user preferences to generate more personalised outfit recommendations.

9.3 AI Virtual Try-On

By mixing apparel and human photos, AI-based virtual try-on (VTO) allows users to see how a particular outfit will look on a person. Recent research has looked into diffusion-based ways to improve the accuracy and realism of virtual try-on outcomes. StableVITON uses latent diffusion and semantic correspondence to transfer clothing onto a target person while maintaining the relationship between the garment and body (Kim et al., 2023).

Other research has focused on preserving the visual details of garments during the try-on process. Texture-Preserving Diffusion Models for High-Fidelity Virtual Try-On aims to maintain important clothing textures and details while generating the final try-on image (Yang et al., 2024). DiffuseFit further addresses challenges such as garment alignment and occluded body regions by combining shape-guided garment warping with diffusion-based image synthesis (AlZubi et al., 2026).

These studies show that diffusion-based algorithms can generate realistic virtual try-on results while addressing issues including garment alignment, texture preservation, and occlusion. Wearly intends to provide an AI virtual try-on tool that allows users to sample selected outfits on their own pictures, enabling another option to explore and assess outfit combinations.

9.4 AI Technologies / Models

Several AI technologies are used in Wearly to support clothing analysis and personalised outfit recommendation.

The **FASHN Human Parser** is used for clothing segmentation, allowing different regions of an uploaded image to be identified and extracted as individual clothing items. This provides the basis for separating clothing items from the original image.

CLIP (Contrastive Language–Image Pre-training) is used for clothing attribute classification, particularly for **colour and style**. By comparing clothing images with predefined descriptive labels, the system can identify relevant visual attributes without requiring a separate classification model.

FashionCLIP is used to extract **fashion-specific visual features** from clothing images. These visual embeddings support clothing similarity analysis and contribute to the outfit recommendation and feedback-based personalisation process.

PyTorch serves as the machine learning framework for loading and running the AI models. Wearly combines these technologies to analyse clothing images, extract relevant visual information, and support AI-assisted outfit recommendations based on clothing compatibility and user preferences.

10.0 Methodology

10.1 Project Approach / Framework

The project will adopt an **Agile development approach** with iterative development and testing. The system will be developed in several iterations, allowing individual modules to be implemented, tested, and improved progressively based on project requirements and supervisor feedback.

The development process will generally follow these stages:

1. Requirement gathering and project planning
2. Literature review and technology evaluation
3. System design and architecture
4. AI model and backend development
5. Mobile application development
6. Integration of AI and mobile application components
7. Testing and evaluation
8. Refinement and finalisation
9. Documentation and presentation preparation

An agile approach is suitable because the AI recommendation and personalisation components may require repeated testing and adjustment throughout development.

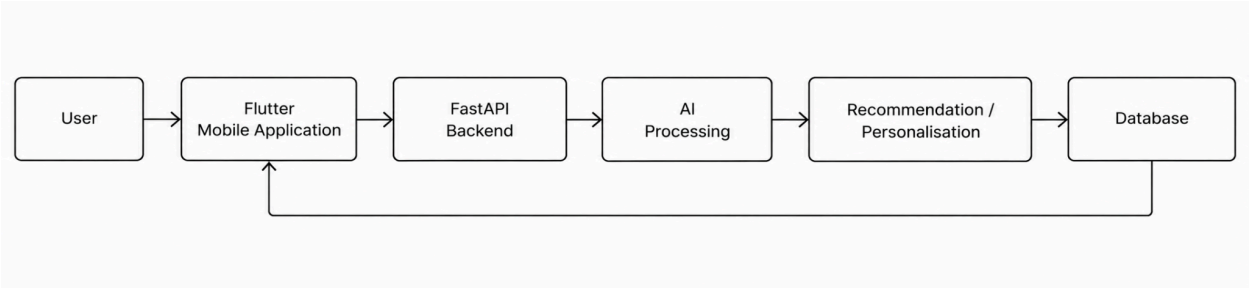
10.2 Tools and Technologies

Category	Technology / Tool	Purpose
Mobile Application	Flutter, Dart	Development of the mobile application
Backend	Python, FastAPI	API development and communication between the application and AI components
Database	MySQL	Storage of user, wardrobe, clothing, and outfit information
Clothing Segmentation	FASHN Human Parser	Extracting clothing items from images
Image Classification	CLIP	Classification of clothing attributes such as colour and style
Visual Feature Extraction	FashionCLIP	Extracting fashion-specific visual features
AI / Machine Learning	PyTorch	Running and integrating AI models
Version Control	GitHub	Source code management and version control

10.3 System Design / Architecture

The proposed system will follow a **client-server architecture** consisting of a Flutter mobile application, FastAPI backend, AI processing components, and MySQL database.

The general system flow will be:



For example, when a user uploads a clothing image, the image will be sent to the backend for AI processing. The system will segment the clothing item, analyse its attributes, extract visual features, and store the resulting information in the user's digital wardrobe. These features will then be used by the outfit recommendation and personalisation components.

10.4 Implementation Plan

The implementation will be divided into the following modules:

Module 1: Digital Wardrobe Management

- Clothing image upload and capture
- Clothing item storage
- Clothing categorisation
- Wardrobe display and management

Module 2: AI Clothing Analysis

- Clothing segmentation using FASHN Human Parser
- Colour and style classification using CLIP
- Fashion-specific visual feature extraction using FashionCLIP

Module 3: AI-Assisted Outfit Recommendation

- Generate compatible combinations of clothing items
- Calculate outfit compatibility based on colour, style, category, and visual features
- Consider previously generated outfits to provide greater variety

Module 4: Personalised Recommendation

- Collect like/dislike feedback
- Automatically build a user preference profile based on clothing attributes and visual features
- Adjust future recommendation rankings based on learned preferences

Module 5: Outfit History

- Record outfits selected or worn by the user
- Display previous outfits through a calendar-based interface
- Use outfit history to reduce repetitive recommendations

Module 6: AI Virtual Try-On

- Allow users to provide a suitable user image for virtual try-on
- Combine the user's image with selected clothing items
- Apply AI-based virtual try-on techniques to generate a visual preview of the selected outfit
- Display the generated result within the mobile application

Module 7: Mobile Application Integration

- Integrate the AI backend with the Flutter application
- Display analysed clothing information
- Display outfit recommendations and personalised results
- Provide interfaces for contextual recommendations and virtual try-on

10.5 Testing and Evaluation

The system will be evaluated through several testing methods:

- **Functional Testing**: To verify that each system function operates according to its requirements.
- **AI Output Evaluation**: To evaluate the accuracy and usefulness of clothing categories, colours, styles, and generated outfit combinations.

- **Recommendation Evaluation**: To determine whether recommended outfits are considered suitable by users and whether personalisation improves recommendations.
- **Usability Testing**: To evaluate ease of use, navigation, clarity of the interface, and overall user experience.
- **Integration Testing**: To ensure that the Flutter application, FastAPI backend, AI components, and database communicate correctly.
- **Performance Testing**: To evaluate response time and system performance during image analysis and outfit generation.

11.0 Expected Deliverables

The expected deliverables of the project are:

- **Functional mobile application** for digital wardrobe management and outfit recommendation.
- **AI-based clothing analysis module** for clothing segmentation, category identification, colour and style classification, and visual feature extraction.
- **AI-assisted outfit recommendation module** for generating compatible outfit combinations.
- **Personalised recommendation module** using like/dislike feedback and user preference learning.
- **Virtual try-on module** for generating visual previews of selected clothing or outfit combinations.
- **Outfit history and calendar module** for tracking previously worn or selected outfits.
- **Backend API** developed using FastAPI for communication between the mobile application and AI components.
- **Database** for storing user, wardrobe, clothing, preference, and outfit history information.
- **System documentation**, including system design, implementation, testing results, and user documentation.
- **Final FYP report and presentation materials.**

12.0 Risk Plan

12.1 Risk Identification and Mitigation

Risk	Potential Impact	Mitigation / Response
Virtual try-on produces inaccurate or unrealistic results	Generated try-on images may not accurately represent the clothing appearance, fit, or alignment on the user	Test the virtual try-on model with different user poses and clothing images and provide suitable image guidelines
AI models produce inaccurate clothing analysis	Incorrect clothing information may affect recommendations	Test different images and improve preprocessing, classification methods, and confidence handling
Outfit recommendations are not suitable	Users may find recommendations inaccurate or unhelpful	Combine multiple factors such as category, colour, style, visual features, and user feedback
AI processing is slow	Poor application responsiveness	Optimise image processing, reduce unnecessary processing, and perform intensive processing on the backend
Integration issues between Flutter and FastAPI	Modules may not communicate correctly	Develop and test APIs independently before full integration
Limited development time	Some planned features may not be completed	Prioritise core functions and treat additional features such as weather, travel, accessories, and hairstyle recommendations as optional extensions

12.2 Risk Assessment

The following risk matrix is used to assess the identified project risks based on their probability of occurrence and potential impact. The combination of probability and impact determines the overall risk priority as low, moderate, high, or critical.

Risk Matrix

Probability ↓ / Impact →	**Low**	**Medium**	**High**
High	Moderate	High	Critical
Medium	Low	Moderate	High
Low	Low	Low	Moderate

Risk Assessment Table

Risk	**Probability**	**Impact**	**Priority**
Virtual try-on produces inaccurate or unrealistic results	High	High	Critical
AI models produce inaccurate clothing analysis	Medium	High	High
Outfit recommendations are not suitable	Medium	High	High
AI processing is slow	Medium	Medium	Moderate
Integration issues between Flutter and FastAPI	Medium	Medium	Moderate
Limited development time	Medium	High	High

13.0 Proposed Gantt chart

WBS NUMBER	TASK TITLE	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	**Project Planning**				
1.1	Requirement Gathering	07/09/26	11/09/26	4	100%
1.2	Project Planning and Scope Definition	14/09/26	18/09/26	4	100%
2	**Literature Review & Technology Selection**				
2.1	Literature Review	18/09/26	21/09/26	3	100%
2.2	AI Model and Technology Evaluation	21/09/26	28/09/26	7	100%
3	**System Design**				
3.1	System Architecture and Database Design	28/09/26	05/10/26	7	100%
3.2	UI/UX and UML Design	05/10/26	12/10/26	7	100%
4	**Mobile Application Development**				
4.1	Flutter UI Development	12/10/26	28/10/26	16	100%
4.2	Feature Integration	23/10/26	29/01/27	96	6%
5	**Backend & AI Development**				
5.1	Backend and Database Development	19/10/26	13/11/26	24	46%
5.2	AI Clothing Analysis	26/10/26	10/11/26	14	29%
5.3	Outfit Recommendation	1/11/26	20/11/26	19	0%
5.4	AI Virtual Try-On	14/12/26	31/12/26	17	0%
6	**Personalisation & Wardrobe Features**				
6.1	Feedback-Based Personalisation	22/11/26	05/12/26	13	0%
6.2	Outfit History & Calendar	06/12/26	14/12/26	8	0%
7	**Extended AI Features**				
7.1	Travel Outfit Recommendation	02/01/27	15/01/27	13	0%
7.2	Accessory Recommendation	18/01/27	22/01/27	4	0%
8	**System Integration & Testing**				
8.1	System Integration	25/01/27	12/02/27	17	0%
8.2	Functional and Integration Testing	08/02/27	19/02/27	11	0%
8.3	Recommendation and Usability Evaluation	15/02/27	05/03/27	20	0%
9	**System Refinement, Documentation**				
9.1	Bug Fixing and System Refinement	01/03/27	15/03/27	14	0%
9.2	Final Report and Documentation	15/03/27	31/03/27	16	0%
10	**Final Presentation & Viva**				
10.1	Final Viva	06/04/27	06/04/27	1	0%



14.0 Resources Required

Hardware

- Laptop for software development and AI processing
- Smartphone for mobile application testing
- Internet connection for downloading libraries, datasets, and development resources

Software

- Flutter SDK and Dart
- Python
- FastAPI
- MySQL
- PyTorch
- Visual Studio Code
- Git and GitHub
- FASHN Human Parser
- CLIP
- FashionCLIP

Datasets / Data

- User-provided clothing images for system testing
- Clothing attribute information generated by the AI models

Other Resources

- Academic research and publications related to artificial intelligence in outfit recommendation and virtual try-on technologies
- Testing participants for usability and recommendation evaluation

15.0 Expected Outcome and Significance

The project is expected to result in a working AI-powered mobile application that allows users to create and manage a digital wardrobe, analyse clothing items with AI, receive personalised outfit recommendations, and preview selected outfits through an AI-based virtual try-on feature.

The system is supposed to show how computer vision and AI-based recommendation approaches may be combined to create a more personalised fashion management experience. The feedback-based personalisation component is also anticipated to improve suggestion relevance over time as it learns from the user's preferences.

From a **technical perspective**, the project demonstrates the integration of pretrained AI models, computer vision, recommendation algorithms, backend APIs, databases, and mobile application development into a single software system.

From a **user perspective**, the system aims to reduce the difficulty of deciding what to wear by providing outfit suggestions based on the user's own wardrobe and preferences. The calendar-based outfit history can also help users keep track of previously worn outfits and explore greater variety.

From a **sustainability perspective**, the project supports SDG 12: Responsible Consumption and Production by encouraging users to make better use of clothing they already own and explore different ways to reuse existing garments. The project also supports SDG 9: Industry, Innovation and Infrastructure through the application of AI and computer vision to develop an innovative digital fashion solution.

The primary beneficiaries are **individuals who often struggle with deciding what to wear and want an easier way to organise their wardrobe and discover suitable outfit combinations**.

16.0 References

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